

When the plan breaks, what happens next?

A recovery workbench for bimanual manipulation.

Shivam Gupta
AI Infra Summit Hackathon

A failure should become a reproducible test

Recurring failure

A cup moves after planning.
A grasp slips during transport.
An obstacle blocks the workspace.

The product response

Configure a bounded task.
Inject a known disturbance.
Inspect the result and export evidence.

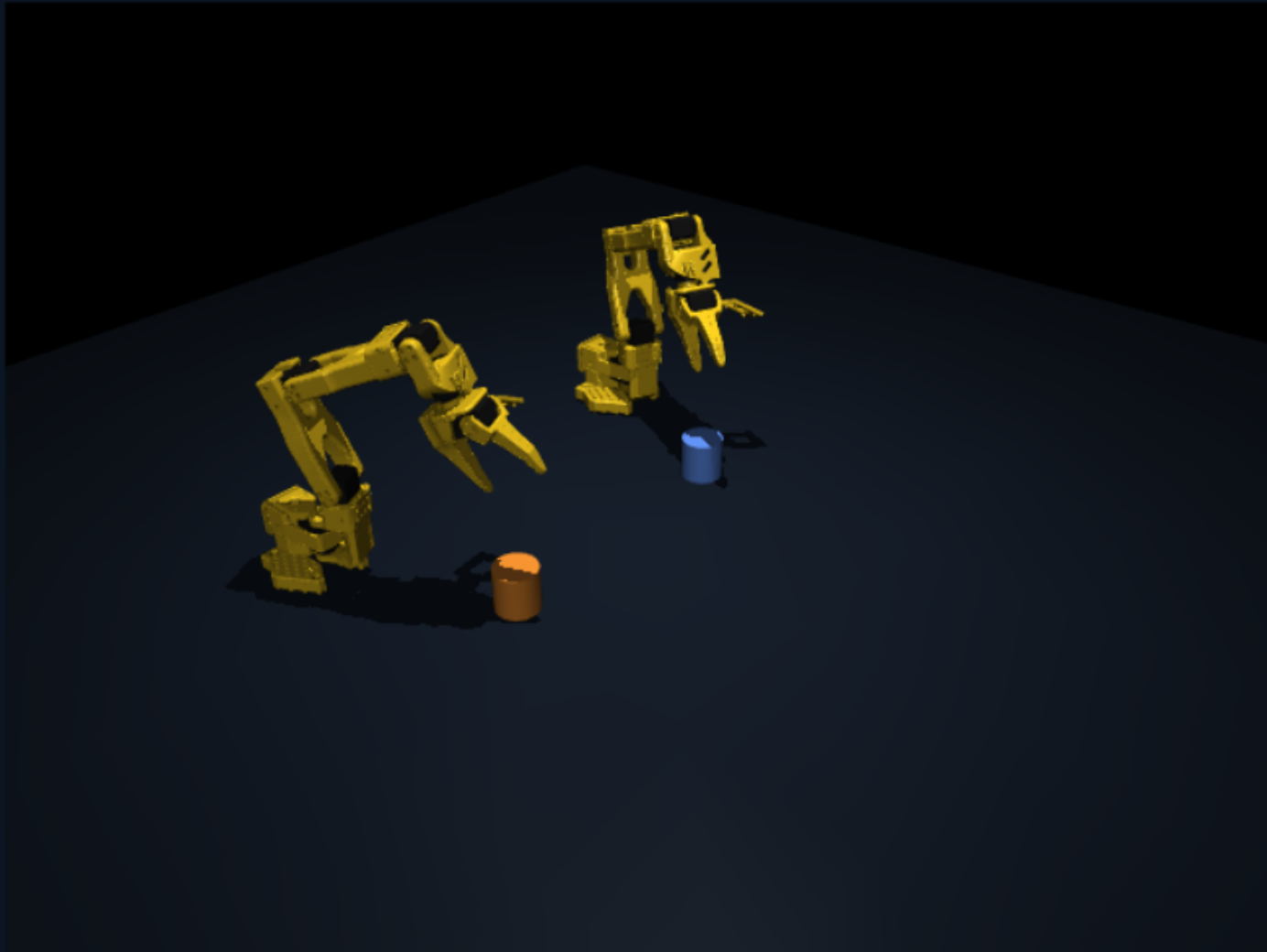
Pilot success means less engineer time per diagnosed regression.

The interface and the physics have separate jobs

Layer	Responsibility
React and Firebase Hosting	Task entry, playback, evidence export, account UI
FastAPI contract	Schema validation, bounded workers, explicit failures
Camera and task validator	Optional image plus instruction, constrained output
MuJoCo and OpenVINO	SO101 control, fault handling, geometric separation
Evidence	Events, sampled states, placement errors, seeded trials

Browser animation illustrates the workflow. Backend evidence records actual simulation.

A bounded contract keeps intent inspectable



Camera + language validator

Raw image, instruction, strict schema

Joint-target imitation policy

480 training / 120 held-out samples

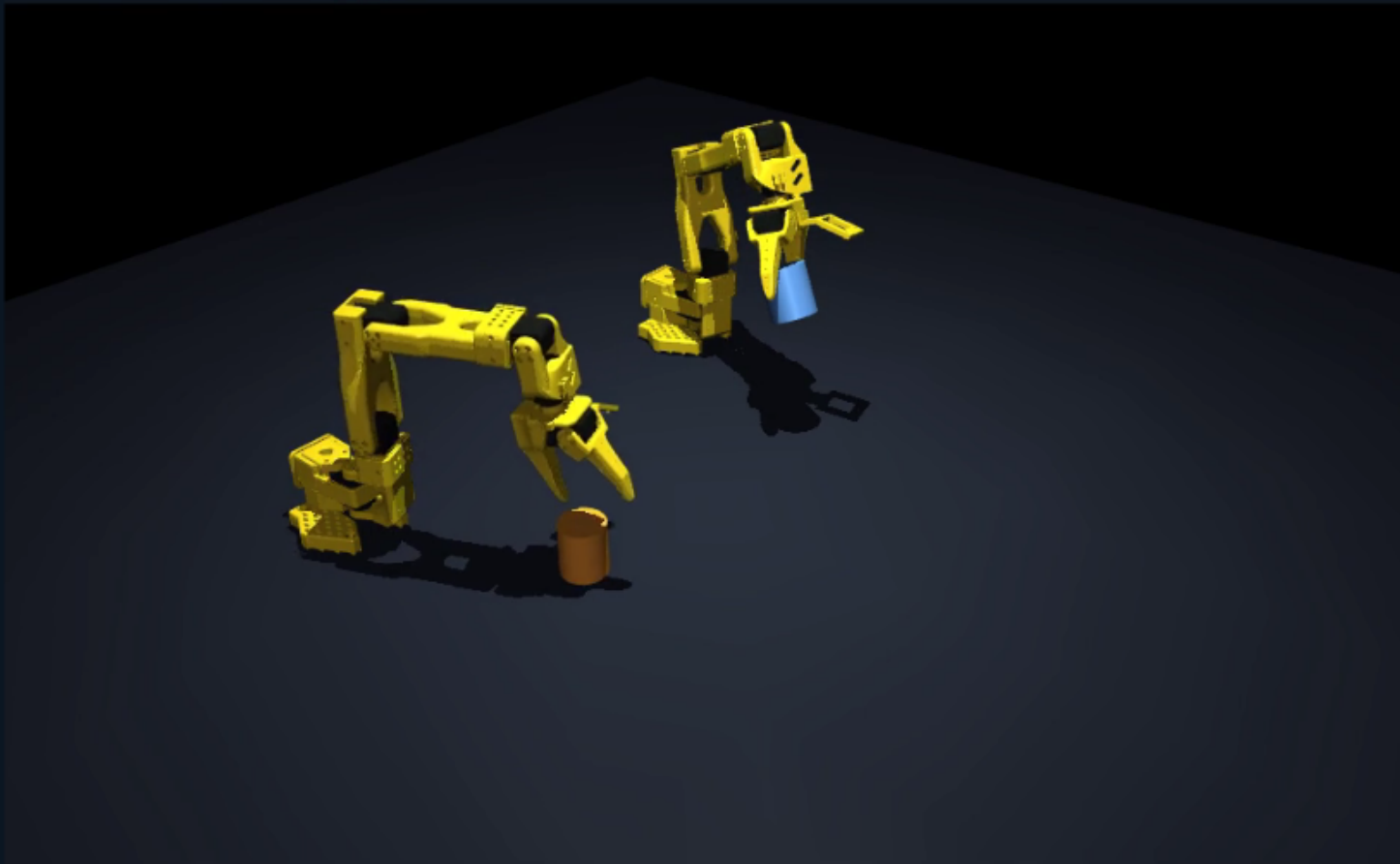
OpenVINO FP32 proposals

Mandatory numerical correction

Held-out joint RMSE: 0.196 rad. State-based policy, not an end-to-end VLA.

Two SO101 arms, with a recoverable grasp fault

GRANTED | Actual MuJoCo dual SO101 rollout
RECOVER 8.00s Fault: grip_loss



Detected cup outside gripper tolerance; paused right arm and regrasped cup

Fault: grip loss

Detect the displaced
object

Regrasp

Verify target error

Video evidence: <backend/evidence/so101-recovery.mp4>

Recovery changed the result in the same fixture

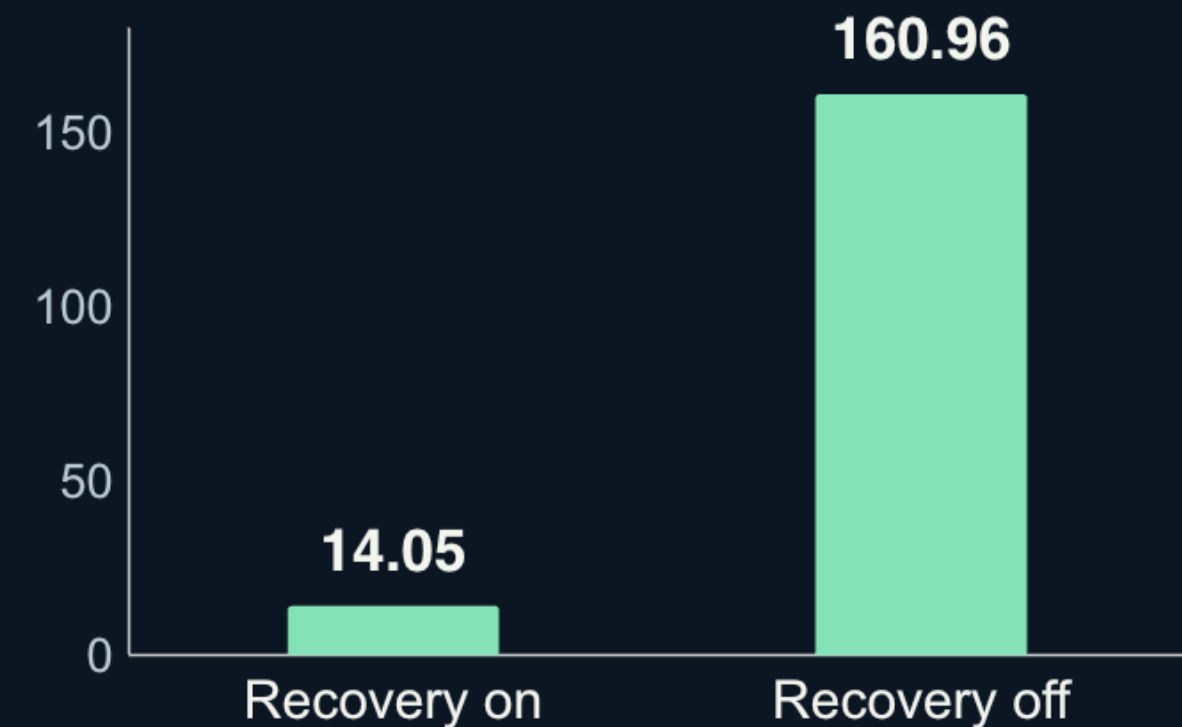
Condition	Completed	Blocked	Mean cup error
Nominal	10 / 10	0 / 10	15.71 mm
Object moved, recovery on	10 / 10	0 / 10	9.09 mm
Grip loss, recovery on	10 / 10	0 / 10	14.05 mm
Grip loss, recovery off	0 / 10	0 / 10	160.96 mm
Obstacle present	0 / 10	10 / 10	169.10 mm

Randomized XY, mass, friction, cylinder size, lighting, and table color.

Every recovered seed retains its measured error

Seed	Cup error	Bowl error
0	15.11 mm	13.16 mm
1	12.73 mm	12.56 mm
2	15.67 mm	12.62 mm
3	15.09 mm	12.67 mm
4	12.81 mm	12.26 mm
5	12.53 mm	12.72 mm
6	16.68 mm	12.97 mm
7	13.72 mm	13.50 mm
8	12.35 mm	12.95 mm
9	13.79 mm	13.11 mm

Mean cup error, mm



Source: backend/evidence/evaluation.json

Learned proposals reduce numerical correction work

Measurement	Result
Policy runtime	OpenVINO CPU, FP32, Apple M4 Pro
Policy latency, median / p95	0.03704 ms / 0.04384 ms
Mean correction iterations	4,528.1 baseline / 4,288.2 learned
Mean local compute time	271.05 ms baseline / 255.76 ms learned
Recovery completion	10/10 baseline and 10/10 learned

No Intel Core Ultra, GPU, NPU, or quantization performance claim.

Evidence maps to the brief, with visible gaps

Track criterion	Points	Current evidence and limit
Task and bimanual manipulation	30	Cup/bowl placement, limited task vocabulary
VLA and multimodal reasoning	20	Camera validator + learned joint proposals, no VLA
Robustness and generalization	15	10 seeds, physics + size + light + color variations
OpenVINO and Intel optimization	20	Policy + graph IR, Intel execution missing
Quality and reproducibility	10	Setup, tests, seeded evaluation, raw artifacts
Innovation and demonstration	5	Visible recovery and an explicit ablation

29 backend tests passed: task rejection, provider failures, safety checks, policy-reference parity.

One integrator. One recurring failure. A measured pilot.

Pilot design

Interview 5 integrators.
Import one existing task.
Reproduce their recurring faults.
Measure time to diagnose a regression.

Pricing hypothesis

\$500 / month per team

\$75 / engineer hour assumed
7 hours saved would cover the fee

No validated demand or revenue. Local execution limits compute costs. Hosted jobs need quotas.

The next release needs stronger evidence

Current boundary	Next verifiable milestone
Fixed two-object task	Complementary manipulation with measurable success
State-based imitation with correction	Vision-language-action policy and held-out tasks
Bounded proxy-object randomization	Novel objects, larger pose shifts, and held-out tasks
Apple development measurements	Intel Core Ultra end-to-end run and device benchmark
Demo infrastructure	Durable storage, tenant isolation, quotas, monitoring

Simulation evidence supports this fixture. It does not certify a physical robot.

**Before the next action,
show the evidence**

granted-robotics.web.app

github.com/shi1720/AI-Infra-Summit-Hackathon

Created by Shivam Gupta
MIT licensed project